GameboyPi – Raspberry Pi Zero 2 W Handheld im Game-Boy-Color-Format

Strom: LiPo 1S (mit Schutz) -> PowerBoost 1000C (Laden + Load-Sharing + 5.2V Boost) -> +5V

Soft-Power: SW9 EIN haelt PWR_REQ low -> Q1 aus -> EN high. SW9 AUS -> GPIO16 low -> Pi faehrt herunter,

danach GPIO4 (gpio-poweroff) low -> Q2 aus -> R8 zieht PWR_REQ high -> Q1 zieht EN low -> Boost aus.

Display: 2.4in ILI9341 SPI an SPI0 (CE0), DC=GPIO25, RST=GPIO24, Backlight an 3V3

Audio: PWM GPIO12/13 -> RC-Filter -> Klinke (Schaltkontakte) -> Mix -> RV1 Lautstaerke -> PAM8302 -> Lautsprecher

Tasten gegen GND, interne Pull-ups: UP=5 DOWN=6 LEFT=19 RIGHT=26 A=23 B=22 START=17 SELECT=27

Akku: ADS1115 AIN0 = VBAT/2 (I2C 0x48), LBO ueber BAT85 an GPIO20, UART-Debug an J8

SW1 BTN_UP GND UP

SW2 BTN_DOWN GND DOWN

SW3 BTN_LEFT GND LEFT

SW4 BTN_RIGHT GND RIGHT

SW5 BTN_A GND A

SW6 BTN_B GND B

SW7 BTN_SELECT GND SELECT

SW8 BTN_START GND START

J1 Raspberry Pi Zero 2 W

+5V (2)			+3V3 (1)
UART_TX	8 GPIO14/TXD	ID_SD/GPIO0	27
UART_RX	10 GPIO15/RXD	ID_SC/GPIO1	28
PWR_SW_SENSE	36 GPIO16	SDA/GPIO2	3 I2C_SDA
BTN_START	11 GPIO17	SCL/GPIO3	5 I2C_SCL
	12 GPIO18/PWM0	GCLK0/GPIO4	7 PWR_KEEP
BTN_LEFT	35 GPIO19/MISO1	GCLK1/GPIO5	29 BTN_UP
LBO_N	38 GPIO20/MOSI1	GCLK2/GPIO6	31 BTN_DOWN
	40 GPIO21/SCLK1	CE1/GPIO7	26
BTN_B	15 GPIO22	CE0/GPIO8	24 LCD_CS
BTN_A	16 GPIO23	MISO0/GPIO9	21 LCD_MISO
LCD_RST	18 GPIO24	MOSI0/GPIO10	19 LCD_MOSI
LCD_DC	22 GPIO25	SCLK0/GPIO11	23 LCD_SCK
BTN_RIGHT	37 GPIO26		
BTN_SELECT	13 GPIO27	PWM0/GPIO12	32 AUD_PWM_L
		PWM1/GPIO13	33 AUD_PWM_R
		GND	6 GND

J2 LCD 2.4in ILI9341 SPI (MSP2401)

1	+5V
2	GND
3	LCD_CS
4	LCD_RST
5	LCD_DC
6	LCD_MOSI
7	LCD_SCK
8	+3V3
9	LCD_MISO
10	
11	
12	
13	
14	

H1 M2.5

H2 M2.5

H3 M2.5

H4 M2.5

H5 Kabel Lautsprecher

H6 M3 Display

H7 M3 Display

J10 Adafruit PowerBoost 1000C

1	VBAT
2	
3	
4	PB_EN
5	GND
6	PB_LBO
7	GND
8	+5V

SW9 POWER: GND (2), PWR_REQ (1), PWR_REQ (2), PWR_SW_SENSE (3)

Q1 BS170: PB_EN, GND, PWR_REQ

Q2 BS170: PWR_REQ, PWR_KEEP, GND

R10 10k: +3V3, LBO_N

R9 100k: VBAT, VBAT_SENSE

R8 100k: VBAT_SENSE, GND

C4 100nF: VBAT_SENSE, GND

R7 100k: VBAT, PWR_REQ

R6 100k: PWR_KEEP, GND

C3 470uF 10V 105°C EKXJ470M6.3X11: +5V, GND

D1 LED rot 3mm: GND, LED_A

D2 BAT85: PB_LBO, LBO_N

R5 270: AUD_PWM_L, AUD_L_F

R4 270: AUD_PWM_R, AUD_R_F

C2 100nF: AUD_L_F, GND

C1 100nF: AUD_R_F, GND

C5 100uF 10V 105°C-A1AKS101: AUD_L_F, HP_L

C6 100uF 10V 105°C-A1AKS101: AUD_R_F, HP_R

J7 SJ1-3525N

S	GND
R	HP_R
RN	SPK_R
T	HP_L
TN	SPK_L

R2 47k: SPK_L, SPK_MIX

R3 47k: SPK_R, SPK_MIX

C7 10nF: SPK_MIX, GND

RV1 PTV09A-2020F-A103 10k log: GND, AMP_IN, SPK_MIX

J8 UART Debug (GND TX RX)

1	GND
2	UART_TX
3	UART_RX

TP1 VBAT, TP2 +5V, TP3 +3V3, TP4 GND, TP5 PB_EN, TP6 AMP_IN

J11 ADS1115 JP2 (VCC..ALERT)

1	+3V3
2	GND
3	I2C_SCL
4	I2C_SDA
5	GND
6	

J12 ADS1115 JP1 (AVDD..AGND)

1	
2	
3	
4	
5	VBAT_SENSE
6	GND

J3 Adafruit PAM8302

1	GND
2	+5V
3	
4	GND
5	AMP_IN

PWR_FLAG +5V, PWR_FLAG GND, PWR_FLAG +3V3, PWR_FLAG VBAT

Raspberry Pi Zero 2 W Handheld

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